Kathryn A. Mummah, PhD

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Education

University of Wisconsin-Madison *Madison, WI*

PhD, Nuclear Engineering and Engineering Physics **2025**

Dissertation: State-Level Nuclear Fuel Cycle Simulations for International Safeguards Applications

MS, Nuclear Engineering and Engineering Physics **2020**

Graduate Certificate in Science Communication, Graduate Certificate in Energy Analysis and Policy

University of Illinois at Urbana-Champaign *Urbana, IL*

Bachelor of Science, Nuclear, Plasma, and Radiological Engineering **May 2017**

Minor in Atmospheric Sciences

Research

Los Alamos National Laboratory *Los Alamos, NM*

R&D Engineer, NEN-5 Nuclear Systems **Mar. 2025 – PRESENT**

- Nuclear fuel cycle systems modeling for international nuclear safeguards, proliferation resistance, and nuclear material control & accounting
- Special purpose reactor design and critical experiment modeling with MCNP
- Sponsors include: NNSA NA-20, NA-80, NRC, DOD/DOW, Industry partners

University of Wisconsin–Madison Dept. of Nuclear Engineering & Eng. Physics *Madison, WI*

Graduate Research Assistant, Computational Nuclear Engineering Research Group (CNERG) **Sept. 2017 – Feb. 2025**

- Improved fidelity of nuclear materials movement modeling within Cyclus for safeguards applications
- Reviewed communication efforts in consent-based siting of radioactive waste disposal facilities
- Designed uranium enrichment cascades in Cyclus to model post-JCPOA cascade performance scenarios
- Sponsors: NNSA NA-22 (university consortia), NA-241

Los Alamos National Laboratory *Los Alamos, NM*

Graduate Research Assistant (GRA), NEN-1 Safeguards Science & Technology **Jan. 2021 – May 2024**

- Developed tool to convert any Cyclus simulation into Code 10-style nuclear material accounting reports
- Generated synthetic advanced nuclear fuel cycles for use with nuclear fuel cycle simulators
- Sponsors: NNSA NA-241

GRA, NEN-2 Advanced Nuclear Technology & Glenn T. Seaborg Summer Research Fellow **May – Aug. 2018**

Visiting Student Guest, NEN-5 Systems Design & Analysis **May – Aug. 2017**

- Conducted acquisition pathway analysis with APAT
- Sponsor: Glenn T. Seaborg Institute

Idaho National Laboratory *Idaho Falls, ID*

Intern, Fuels Modeling & Simulation **May – Aug. 2016**

- Benchmarked pellet-cladding mechanical interactions in BISON

Selected Honors & Awards

2025: *Nuclear News* 40 Under 40
2019-24: *Graduate Fellow*, Nuclear Nonproliferation International Safeguards Program
2022: ANS Presidential Citation
2022: Best Presentation in Nuclear Technology & Society, ANS 2022 Student Conference
2020: First Place, Institute of Nuclear Materials Management J.D. Williams Student Paper Award
2016: ANS Student Sections Committee Commendation for Service and Leadership

Leadership & Service

American Nuclear Society (ANS) *Member since 2013*

- Chair, Decommissioning & Environmental Sci, 2025-27
- Chair, Student Sections Committee, 2021-24
- Exec Comm, Fuel Cycle & Waste Management, 2021-24
- Student Director, ANS Board of Directors, 2017-19
- Exec Comm, Young Members Group, 2019-22
- Exec Comm, Nuclear Nonproliferation Policy, 2018-21